



ARTI 406 - Machine Learning

Assignment 2: Data Preprocessing

Dataset: Video Game Sales (vgsales.csv)

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Task 1:

	Missing Count	Missing (%)
Year	271	1.63
Publisher	58	0.35

Figure 1: Bar chart (missing values %)

Duplicate rows: 0

Figure 2: Check Duplicates

```
Rank          int64
Name          object
Platform      object
Year          float64
Genre         object
Publisher      object
NA_Sales      float64
EU_Sales      float64
JP_Sales      float64
Other_Sales   float64
Global_Sales  float64
dtype: object
```

Figure 3: Check Data Types & Basic Stats

	Rank	Year	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales
count	16598.000000	16327.000000	16598.000000	16598.000000	16598.000000	16598.000000	16598.000000
mean	8300.605254	2006.406443	0.264667	0.146652	0.077782	0.048063	0.537441
std	4791.853933	5.828981	0.816683	0.505351	0.309291	0.188588	1.555028
min	1.000000	1980.000000	0.000000	0.000000	0.000000	0.000000	0.010000
25%	4151.250000	2003.000000	0.000000	0.000000	0.000000	0.000000	0.060000
50%	8300.500000	2007.000000	0.080000	0.020000	0.000000	0.010000	0.170000
75%	12449.750000	2010.000000	0.240000	0.110000	0.040000	0.040000	0.470000
max	16600.000000	2020.000000	41.490000	29.020000	10.220000	10.570000	82.740000

Figure 4: df.describe() table

Task 2:



```
Shape before: (16598, 11)
Shape after: (16327, 11)

Remaining missing values:
Rank      0
Name      0
Platform  0
Year      0
Genre     0
Publisher 0
NA_Sales  0
EU_Sales  0
JP_Sales  0
Other_Sales 0
Global_Sales 0
dtype: int64
```

Figure 5: Missing Value Strategy

Task 3:

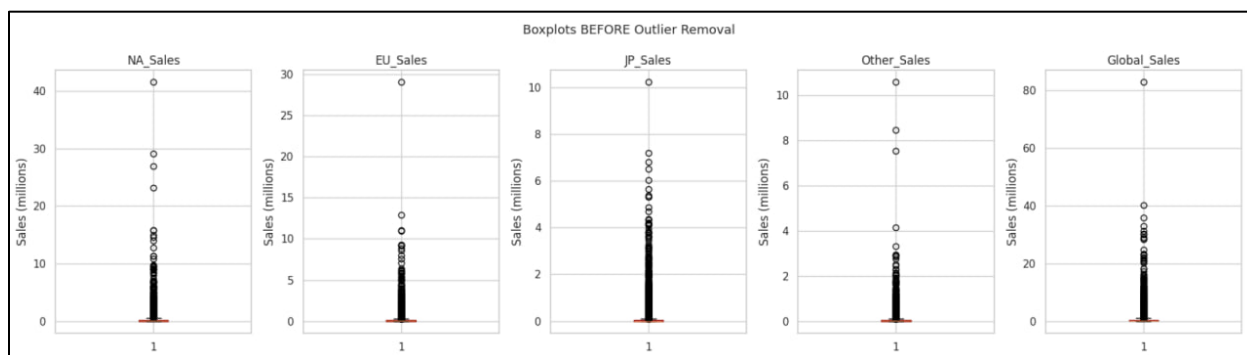


Figure 6: Boxplots BEFORE

```
NA_Sales: Q1=0.000, Q3=0.240, IQR=0.240, Lower=-0.360, Upper=0.600, Outliers=1660
EU_Sales: Q1=0.000, Q3=0.110, IQR=0.110, Lower=-0.165, Upper=0.275, Outliers=2054
JP_Sales: Q1=0.000, Q3=0.040, IQR=0.040, Lower=-0.060, Upper=0.100, Outliers=2407
Other_Sales: Q1=0.000, Q3=0.040, IQR=0.040, Lower=-0.060, Upper=0.100, Outliers=1645
Global_Sales: Q1=0.060, Q3=0.480, IQR=0.420, Lower=-0.570, Upper=1.110, Outliers=1827

Rows before: 16,327 → After: 11,801
```

Figure 7: IQR bounds and remove outliers

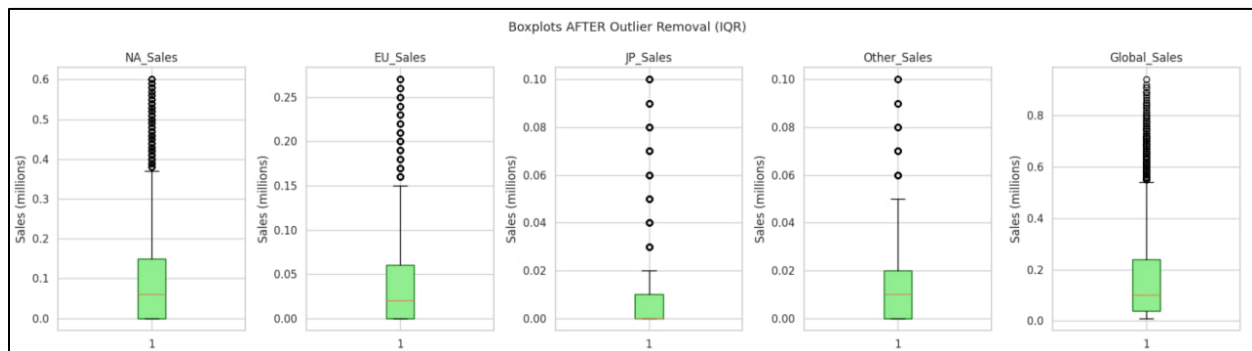


Figure 8: Boxplots AFTER

Task 4:

Min-Max (first 5 rows):				
	NA_Sales_minmax	EU_Sales_minmax	JP_Sales_minmax	Other_Sales_minmax \
0	0.850000	0.962963		0.9
1	0.966667	0.962963		0.0
2	0.966667	0.888889		0.1
3	0.950000	0.962963		0.0
4	1.000000	0.814815		0.5
Global_Sales_minmax				
0	1.000000			
1	0.978495			
2	0.967742			
3	0.967742			
4	0.946237			
Z-score (first 5 rows):				
	NA_Sales_zscore	EU_Sales_zscore	JP_Sales_zscore	Other_Sales_zscore \
0	3.381558	3.712352	3.574521	3.619377
1	3.960430	3.712352	-0.467968	3.619377
2	3.960430	3.372449	-0.018802	4.160450
3	3.877734	3.712352	-0.467968	3.619377
4	4.125822	3.032545	1.777859	0.372942
Global_Sales_zscore				
0	4.608484			
1	4.489312			
2	4.429726			
3	4.429726			
4	4.310554			

Figure 9: Min-Max first 5 rows table & Z-score first 5 rows table

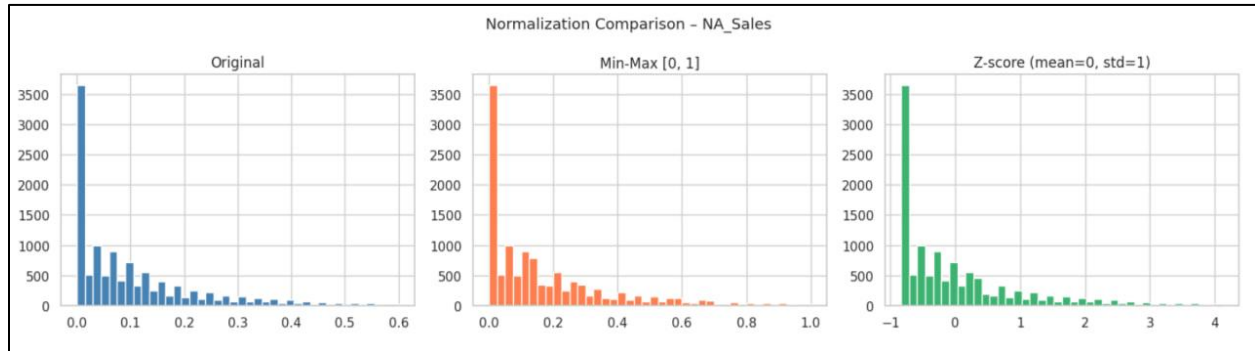


Figure 10: 3 histograms comparison

Task 5:

PC	Explained Var (%)	Cumulative (%)
PC1	64.53	64.53
PC2	19.14	83.68
PC3	11.58	95.25
PC4	4.74	99.99
PC5	0.01	100.00

Figure 11: PC table

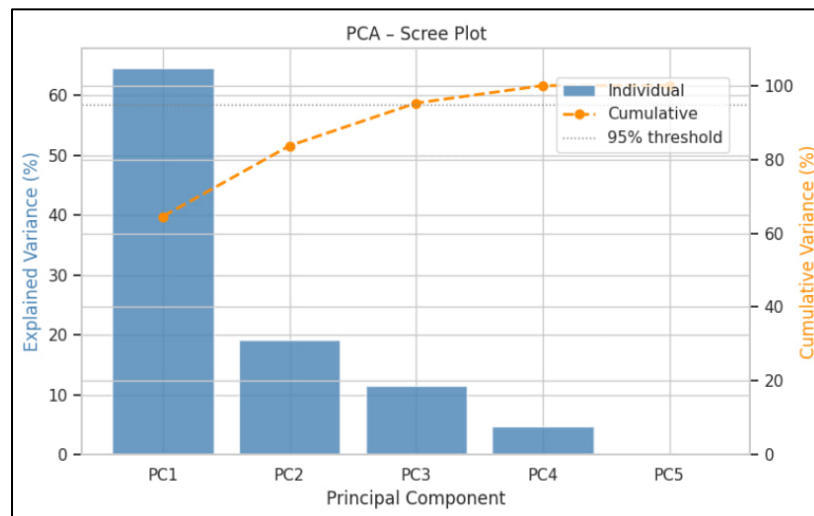


Figure 12: Scree plot



	PC1	PC2	PC3	PC4	PC5
NA_Sales	0.4768	-0.0403	0.6753	-0.0600	-0.5581
EU_Sales	0.4673	0.0715	-0.6028	-0.5821	-0.2727
JP_Sales	-0.1580	0.9778	0.0820	-0.0441	-0.1016
Other_Sales	0.4886	0.1342	-0.3330	0.7912	-0.0803
Global_Sales	0.5391	0.1386	0.2511	-0.1723	0.7729

Figure 13: Loadings table